

جامعة أم القرى
UMM AL-QURA UNIVERSITY

Course: Software Engineering Fundamentals

Project Title: Simple Budget Planner
Instructor: Dr. Basmah M. Alsenan

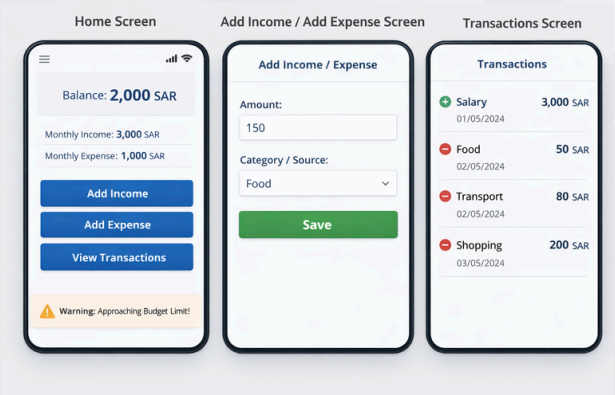
Ghala Hussain Alsharif
Ghadi Bandar Mukhtar
Yara Shadi Al Ansari
Jood Abdullah Al Osaimi
Anhar Mohammed Al Nafiei

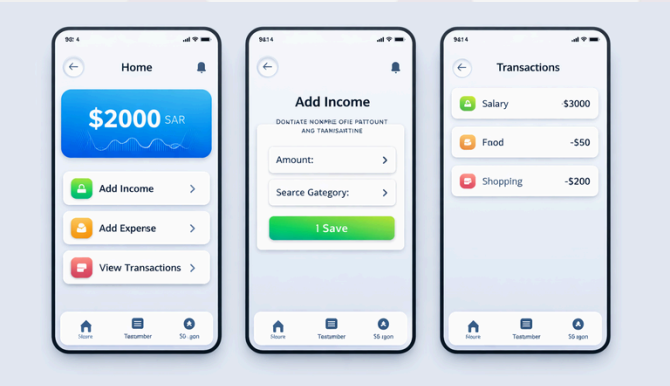
Part 4

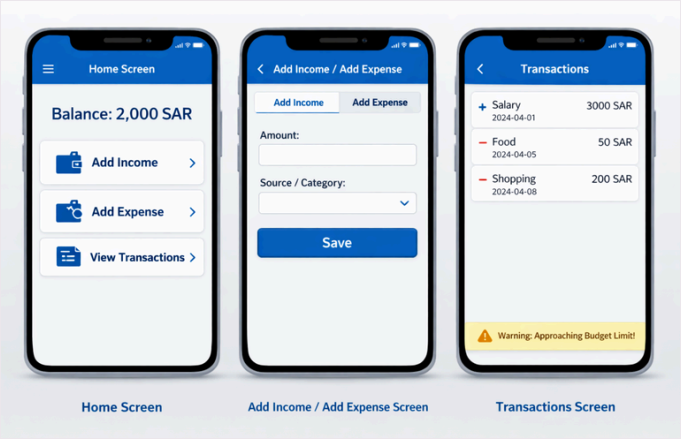
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Prototype

This is a High-Fidelity Prototype because it shows realistic user interfaces and system interactions.







Implemented Parts: Home Screen, Add Income/Add Expense Screen, Transactions Screen, Budget Alert Feature.

Importance of the Prototype:

Helps validate the system's core functions early.

Allows users to test the interface before full development.

Ensures the design meets the needs of students and young employees.

Provides a foundation for adding advanced features later.

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Test cases

Test Case ID	Test Description	Test Steps	Expected Result	Actual Result	Status
TC1	Add Income	Enter amount & source, click Save	Saved & balance updated	As expected	Pass
TC2	Add Expense	Enter amount & category, click Save	Saved & balance reduced	As expected	Pass
TC3	View Transactions	Open Transactions screen	All details displayed	As expected	Pass
TC4	Balance Calculation	Add transactions, check Home	Correct remaining balance	As expected	Pass
TC5	Budget Alert	Add expense near limit	Warning notification	As expected	Pass

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Testing Results

The system was tested using the planned test cases. Overall, the prototype worked as expected and successfully performed its main functions.

Results:

- Income was added successfully and the balance was updated correctly.
- Expenses were recorded successfully and deducted from the balance.
- Transactions were displayed correctly in the transactions screen.
- The monthly budget was saved successfully.
- Budget alerts appeared when spending approached the budget limit.

Problems Found:

- Minor delays occurred when moving between some screens.
- Alert messages could be improved to be more noticeable.

Conclusion:

The prototype achieved its objectives and demonstrated the main functionality of the Simple Budget Planner system successfully.

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System Limitations

• The system does not support direct integration with bank accounts or credit cards.
• Users must manually enter income and expense data.
• The system does not provide online payment or money transfer services.
• The system does not include investment or stock market features.
• The accuracy of the system depends on user-entered data.
• The system does not provide AI-based financial predictions or recommendations.

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Reflection

We learned a lot from the Simple Budget Planner project. It helped us understand the software engineering process better. This process includes analyzing problems, gathering requirements designing the system and creating the user interface. The project showed us how software can solve problems with personal finance.

One big challenge was keeping all parts of the system consistent. We had to ensure that the requirements, use cases, system design, UML diagram and user interface all worked towards the goals.

In the future we can improve the system by adding features. Some ideas are financial reports, charts, data analysis and better security. We could also turn the prototype into an application with a more interactive user experience.

Overall the project gave us experience. We learned how to apply software engineering concepts to create an easy-, to-use system. The Simple Budget Planner project was a learning experience.

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Student	Task Completed
Ghadi Bandar Mukhtar	Prototype
Yara Shadi Al Ansari	Testing Results
Ghala Hussein Al Sharif	Reflection
Joud Abdullah Al osaime	Test Cases
Anhar mohammed al nafiei	System Limitations



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